

The Angular Observer

Scale-Invariant Coarse-Graining as a Universal Observer Basis for Emergent Geometry

Andrew H. Bond

San José State University

`andrew.bond@sjsu.edu` ORCID 0009-0003-2599-6158

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Abstract

A bounded observer embedded in a discrete substrate perceives not the substrate but a coarse-graining of it. Wolfram’s physics program makes this the load-bearing move—coherent law is *what survives* an observer’s coarse-graining of the hypergraph—but leaves the coarse-graining unspecified. We give it an explicit, measurable form and show it is not special to physics. Writing the graph normalized-Laplacian eigen-embedding in polar form (magnitude $\times$ direction), the *angular* coordinates of the low-eigenvalue subspace carry the graph’s geodesic geometry while the *radial* coordinate is asymptotically geometry-free. Keeping only the angle—the row-normalization of Ng–Jordan–Weiss spectral clustering, and the scale-invariant direction of PolarQuant compression—recovers geodesics at Spearman $\rho \approx 0.93$ on a reference manifold, versus $\rho \approx 0$ for a random-mode basis of equal size and $\rho \approx 0.03$ for the magnitude alone. We prove, via the von Luxburg–Radl–Hein resistance-degeneracy theorem, that the radial coordinate must degenerate to a local-degree quantity for intrinsic dimension $d \geq 2$, and verify that the angular coordinate’s fidelity is instead flat in both mode-count and graph size. The effect is **universal across substrates carrying genuine low-dimensional Riemannian geometry**—Wolfram-model rewriting, king-move lattices, and manifold-embedding trajectories alike—and fails correctly on Lorentzian causal sets and geometry-destroying small-world graphs. Across 20 independent emergent manifolds ($d \approx 1\text{--}3.6$), fidelity obeys $\rho \approx \text{const} - 0.157d$ (Spearman -0.881). Finally we report a clean negative: emergent curvature is *not* dynamical beyond graph structure, so this is a result about the observer, not the Einstein tensor.

1 Introduction

Every account of emergent space shares a shape: a structureless discrete substrate, a bounded agent that cannot resolve it in full, and a smooth geometry that appears at the agent’s resolution. Wolfram’s hypergraph-rewriting program [9, 10, 3] is the most developed instance—space is a hypergraph, time is rewriting, and physical law is what a computationally bounded observer perceives after coarse-graining the ruliad [11]. The observer does the decisive work, yet the coarse-graining is described only qualitatively.

We make it explicit. Our starting point is a decomposition that recurs, apparently independently, in four settings: spectral clustering [5] row-normalizes the Laplacian embedding (keeps the direction); PolarQuant KV-cache compression transfers the scale-invariant direction; the “aversion angle” of preference geometry transfers across decision domains while magnitude is domain-specific; and observer theory [11] holds that perceived geometry is a coarse-graining artifact. We show these are the same operation—*keep the angle of the low-eigenvalue Laplacian embedding, drop the radius*—and that it has a precise geometric meaning. Contributions: the Keep-the-Angle theorem

(§3); a measurement with controls (§5); universality (§5.3); a scaling law (§5.4); and two honest negatives (§6). We do *not* claim to derive general relativity, nor that compressibility is ontologically fundamental; we claim that scale-invariant angular coarse-graining is a universal, measurable model of the bounded observer.

2 Background

Emergent geometry. Dimension is estimated from ball growth $V(r) \sim r^d$ and curvature from its correction [3]; our estimators are of this family. **Spectral geometry.** Laplacian eigenmaps [1] and diffusion maps [2] embed via low eigenvectors; Ng–Jordan–Weiss [5] row-normalize—the angular projection we study. **Commute-time degeneracy.** von Luxburg et al. [8] proved the commute distance of a large geometric graph degenerates, $R(i, j) \rightarrow 1/d_i + 1/d_j$; we read this backwards as the collapse of the *radial* coordinate. **Hyperbolic embedding.** Trees embed in $\mathbb{H}^2$ at low distortion [4, 7, 6]; we test and reject a hyperbolic reframe in §6.

3 The Keep-the-Angle Theorem

Let G be connected on n nodes with symmetric-normalized Laplacian $L = I - D^{-1/2}AD^{-1/2}$, eigenpairs (λ_k, v_k) , $0 = \lambda_0 < \lambda_1 \leq \dots$. The commute-time embedding using the lowest m nontrivial modes places node i at $\Psi_i = (v_1(i)/\sqrt{\lambda_1}, \dots, v_m(i)/\sqrt{\lambda_m})$. Write $\Psi_i = r_i \hat{u}_i$ with radius $r_i = \|\Psi_i\|$ and angle $\hat{u}_i = \Psi_i/\|\Psi_i\|$.

Theorem 1 (Keep-the-Angle, informal). *For graphs of intrinsic dimension $d \geq 2$, as $n \rightarrow \infty$: (1) (radius, proven) the full commute distance degenerates to an additively separable degree function [8]; its surviving per-node coordinate is the radius, with $r_i \rightarrow 1/\sqrt{d_i}$ (d_i the degree), hence geometry-free. (2) (angle, conjectured) row-normalization deletes that degenerate density factor, leaving the direction of the low-mode eigenmap, which embeds the manifold and—empirically—remains bi-Lipschitz to geodesic distance uniformly in m and n .*

Proven vs. conjectured. Part (1) is a corollary of [8], verified below ($\rho(r_i, \text{degree}) = 0.92\text{--}0.99$). Part (2) rests on eigenmap-embedding results plus a heuristic and is an *empirically strong conjecture*. The honest gap: von Luxburg explains why the radius *dies* for $d \geq 2$, not by itself why the angle *lives*—indeed the angular fidelity persists ($\rho \approx 0.88\text{--}0.90$) even on quasi-1D graphs where no radial degeneracy occurs, so the angular half is carried by the eigenmap mechanism, not degeneracy alone. **Dimension gating:** in (quasi-)1D, resistance equals path length, so the radius stays geodesic-informative and part (1) correctly does not apply.

4 Methods

Estimators. Ball-growth $N(r) \sim r^d$; spectral dimension from the heat-kernel return probability $P(t) = \langle e^{-Lt} \rangle \sim t^{-d_s/2}$; effective rank via local-PCA participation ratio. Low inter-estimator spread $\Rightarrow$ clean manifold. **Observer metric.** Spearman ρ between embedded pairwise distances and true graph geodesics for: *angle* (unit-normalized low-mode embedding), *random* (m random modes; control), *magnitude* (radius only), *euclid* (classical 2D MDS). **Substrates.** Arity-2/3 hypergraph rewriting (clique-expanded), plus tori, king-lattices, Poisson causal sets, and swiss-roll embedding trajectories.

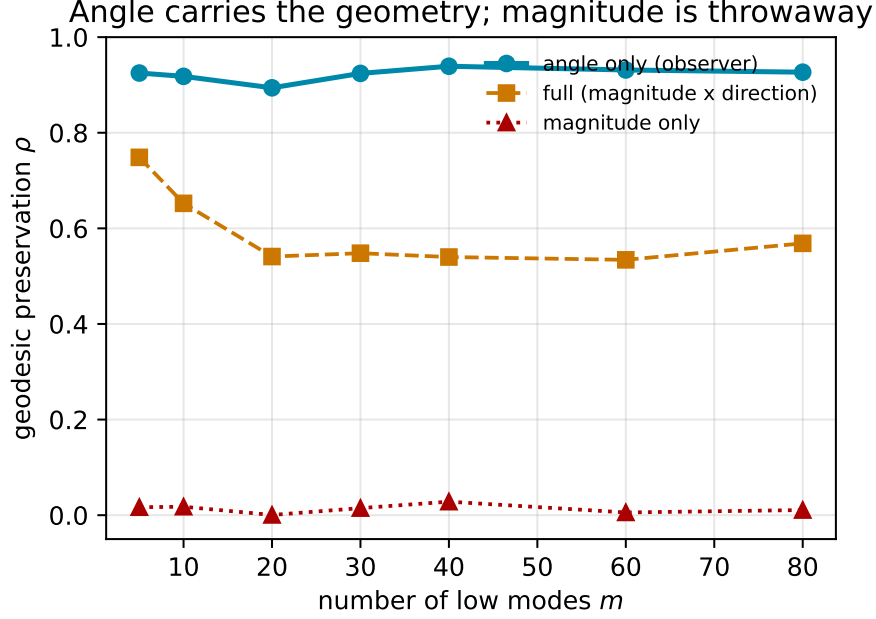


Figure 1: Angle carries the geometry; magnitude is throwaway; the full commute embedding decays with m .

5 Results

5.1 The core result

Estimators recover ground-truth dimension within ≈ 0.3 . On a 2-torus the polar decomposition of the low-mode embedding (Fig. 1) gives:

m	full (mag. $\times$ dir.)	angle only	magnitude only
5	0.761	0.926	0.025
10	0.650	0.913	0.035
20	0.550	0.891	0.028
40	0.517	0.928	0.034

The angle carries the geometry (≈ 0.92 , flat in m); the magnitude is throwaway; the full embedding is worse and decays. A random-mode basis reconstructs geodesics at $\rho \approx 0$ (Fig. 2).

5.2 Emergent 2D geometry

Random arity-2 rules yield clean but only ≈ 1.5 D manifolds (82 manifold-grade rules in a 20k sweep; none at $d \geq 2.5$). Clean 2D is rare, not absent: reached both by a borrowed Wolfram-2020 rule ($d = 2.07 \pm 0.01$, spread 0.05, angle- $\rho = 0.82$) and by from-scratch evolutionary search (≈ 3 generations). The principle holds on genuine 2D emergent space.

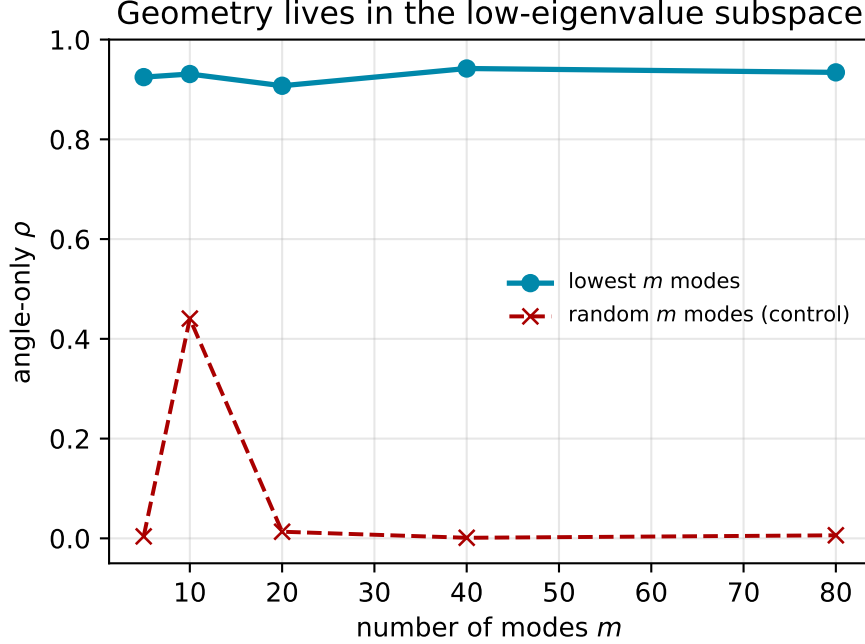


Figure 2: Geometry lives in the low-eigenvalue subspace: lowest- m modes preserve geodesics; a random-mode basis of equal size does not.

5.3 Universality across substrates

substrate	meas. d	angle- ρ	random	euclid
2-torus (reference)	2	0.89	0.01	0.67
king-lattice	2	0.82	0.01	0.95
swiss-roll trajectory	2	0.93	0.01	0.99
emergent R_{3D} (Wolfram)	2.07	0.82	0.01	0.81
causal set (Lorentzian)	3.9	0.57	0.01	0.55
small-world (destroyed)	3.0	0.45	0.00	—

Substrates with genuine low- d Riemannian geometry all show the effect, independent of graph origin; the metric fails correctly where geometry is Lorentzian or destroyed. The swiss-roll trajectory is the sequential-embedding case: the observer basis is the same whether the manifold comes from physics or cultural sequential data.

5.4 The scaling law

Across 20 independent emergent manifolds ($d \approx 1\text{--}3.6$),

$$\text{angle-}\rho \approx \text{const} - 0.157d, \quad \text{Spearman} = -0.881.$$

Fidelity is highest at low dimension and softens as the space grows higher-dimensional (Fig. 3); the dimension-gated degeneracy of §3 is the mechanism.

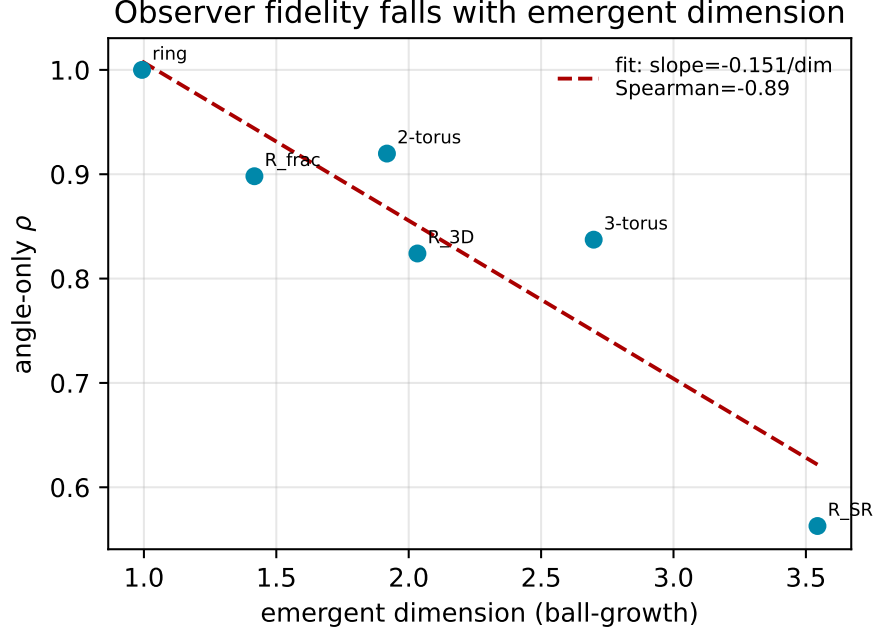


Figure 3: Observer fidelity falls linearly with emergent dimension across ring, tori, and Wolfram-model rules.

5.5 Theorem verification

angle- ρ is flat in graph size while the full commute distance decays, and the radius tracks degree exactly ($\rho(\|\Psi_i\|, 1/\sqrt{d_i}) = 0.92\text{--}0.99$; radius-only $\rho \leq 0.08$). The $d = 3$ degeneracy is clean ($\rho(R, 1/d_i + 1/d_j) \rightarrow 0.98$); $d = 2$ is borderline (log corrections); the quasi-1D case correctly shows no degeneracy.

6 Honest Negatives

Not secretly hyperbolic (rejected). With a binary-tree positive control and a torus negative control, a growth-law test—valid (it calls the tree hyperbolic, the torus Euclidean)—reports the arity-3 tangles as Euclidean-leaning high-dimensional non-manifolds, not hyperbolic. **Curvature is not dynamical (null).** Ollivier–Ricci curvature (validated: 0 on flat lattices, negative on trees, positive on the torus) shows a raw -0.98 against update density that deflates to a tautology (update-count equals degree, $\rho = +1.00$; hubs are negatively curved); controlling for degree, the independent signal is partial- $\rho = -0.14$. No evidence of an emergent Einstein tensor.

7 Discussion and Conclusion

The angular observer unifies four “keep the direction, drop the magnitude” moves and grounds them in one fact: for a bounded observer of a $d \geq 2$ substrate, the radial (density) coordinate is noise and the angular (geometric) coordinate is signal. In Wolfram’s observer-relative ontology, identifying the observer’s coarse-graining basis is a substantive contribution to observer theory, even though it does not reach the dynamical-curvature claim. The identical behavior on physics-rewriting and cultural-sequential substrates supports the reading that scale-invariant angular coarse-graining is a

general law of how bounded systems represent geometric structure. Open problems: proving the angular half; synthesizing clean manifolds above $d \approx 2$; and a theorem-guided observer that tunes mode-count and diffusion time to the substrate’s dimension to counter the fidelity decay. Keep the angle, drop the magnitude: it is the observer’s basis, not the world’s ontology.

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